

Estimating False Multistation N2 Upstaging From Cross-Contamination During EBUS-TBNA Without ROSE

Insights From a Monte Carlo Analysis

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Background: The ninth TNM edition distinguishes single-station (N2a) from multistation (N2b) disease, increasing concern for cross-contamination between N2 lymph nodes during EBUS staging when ROSE is unavailable. The absolute probability and clinical impact of this phenomenon are uncertain. We estimated a theoretical upper bound on false upstaging and the number of needle changes required to prevent one false upstage.

Methods: A Monte Carlo decision model simulated mediastinal staging in 100,000 virtual patients across published estimates of occult N2 prevalence, probability of a single-station involvement, and contamination rates. Outcomes were the probability of false N2b upstaging and the number of additional needle changes needed to prevent one false upstage (NNC). Sensitivity analyses varied the contamination rate, single-station probability, and sampling order.

Results: False upstaging in central cN0 disease was ~0.5% to 0.8% in the base case, increasing to 1.4% in higher-risk scenarios. cN1 disease showed slightly higher probabilities. Needle-change efficiency was low; preventing one false upstage required roughly 250 to 350 extra needle changes in central cN0 and >1000 in peripheral cN0. Sensitivity analysis showed comparable influence of contamination rate, single-station probability, and sampling order.

Conclusion: The risk of spurious upstaging during EBUS staging without ROSE is small but not negligible, concentrated in central cN0 and cN1 tumors. When ROSE is not available, routine needle change between N2 stations provides minimal benefit for most patients. In cases where ROSE is not available and resources are limited, a selective strategy targeting high-risk contexts may offer the best balance between accuracy and procedural efficiency.

Key Words: EBUS, lung cancer, cross contamination, Monte Carlo, staging

(*J Bronchol Intervent Pulmonol* 2026;33:e01070)

Accurate mediastinal staging is a cornerstone of treatment planning in non-small cell lung cancer (NSCLC). Endobronchial ultrasound-guided transbronchial needle aspiration (EBUS-TBNA) has largely replaced mediastinoscopy as the first-line modality for sampling mediastinal and hilar lymph nodes. Bronchoscopists are trained to sample nodes in descending order of stage—from the highest-risk station (N3), to N2, and then N1.¹ The logic is straightforward: if a higher-stage node was positive, inadvertent contamination of lower-stage nodes would not affect the final stage. In addition, a positive lymph node for malignancy on rapid onsite evaluation (ROSE) would negate the need to sample the lower-stage lymph nodes. Under the seventh and eighth editions of the TNM classification, this convention minimized the risk of false upstaging without practical consequence for patient management even when ROSE is not available.

The ninth edition of the TNM classification introduced an important refinement: subdivision of N2 disease into single-station (N2a) and multistation (N2b) involvement.² This distinction carries both prognostic and potential therapeutic implications. Patients with N2a disease may be considered for surgical resection in selected cases, although treatment strategies for N2 disease frequently overlap and depend on physiological reserve and multidisciplinary assessment. In contrast, N2b disease is generally excluded from surgical management. This shift means that what was once clinically irrelevant potential contamination—seeding malignant cells from one N2 station into another—may now result in a meaningful change in stage and treatment.

The use of ROSE may improve sample adequacy, reduce the need for additional biopsies, and lower complication rates and procedure time though its diagnostic accuracy varies by center.^{3–6} However, ROSE availability varies considerably across institutions. In a recent U.S. survey, ~12% of pulmonologists reported no access to ROSE, a figure likely higher in community practices and in low-resource international settings.⁷ The absence of ROSE during EBUS for mediastinal staging may increase the potential for a malignant station sampled early to contaminate a negative station sampled later. Prior EUS and EBUS studies have demonstrated residual malignant cells within the needle or channel after sampling a

Received for publication December 4, 2025; accepted February 26, 2026.

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IRB exemption was obtained through our institution (King Faisal Specialist Hospital and Research Centre) with an ID number of 13178. A.S., S.M., I.A., S.A., and L.A.: conception and design, acquisition of data, drafting the article, critical revision of intellectual content, and final approval of the version to be published. The manuscript has been read and approved by all the authors. The requirements for authorship as stated earlier in this document have been met, and each author believes that the manuscript represents honest work.

There is no conflict of interest or other disclosures.

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Supplemental Digital Content is available for this article. Direct URL citations are provided in the HTML and PDF versions of this article on the journal's website, www.bronchology.com.

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DOI: 10.1097/LBR.0000000000001070

malignant node, suggesting that microscopic contamination can occur.^{8,9}

Contemporary risk models including HAL (Help with Assessment of Adenopathy in Lung Cancer) and HOMER (Help with Oncologic Mediastinal Evaluation for Radiation) indicate that ~2% to 4% of patients with peripheral PET/CT-negative tumors, around 7% of those with central PET/CT-negative tumors, and roughly 27% of those with cN1 disease harbor unsuspected N2 metastases.¹⁰ Patient's age, histology type, N stage by CT scan, N stage by PET as well as location of the tumor (central vs. peripheral) were used in the model to predict the risk of N2/N3 disease.

The prevalence of N2a vs N2b disease have been studied. In patients with radiographic N0 disease, the prevalence of pathologic N2a ranges from 4.1% to 6.5%, while pathologic N2b ranges from 1.2% to 2.4%. In patients with radiographic N1 disease, the prevalence is higher for both categories but maintains a similar ratio: 18.5% had pN2a and 14.8% had pN2b.¹¹ Other studies found similar results with a prevalence of N2a around 60% to 70%.^{12,13} These estimates reflect pathologic N2 distributions and not strictly the subset of clinically occult N2 disease. Therefore, the true distribution among clinically unsuspected cases remains uncertain. In the era of TNM-9, these distributions matter greatly: if single-station disease is more common, the risk of inadvertent upstaging by contamination also becomes more relevant. This is important, as single-station N2 disease has a superior prognosis compared with multistation N2 disease, and mislabeling a patient as such may potentially affect treatment decisions.^{13,14}

Against this backdrop, a practical and unanswered question arises: in the absence of ROSE, should bronchoscopists routinely change the needle between sampling N2 stations to prevent false upstaging? Or is the absolute risk so small that the cost and procedural burden outweigh the benefit?

Monte Carlo simulation has previously been applied in thoracic oncology to address procedural strategy when randomized trials are impractical. A 2022 analysis in CHEST modeled whether diagnostic workup should begin with navigation bronchoscopy or CT-guided biopsy and demonstrated the value of simulation in quantifying procedural trade-offs under uncertainty.^{15,16} This provides a clear precedent for applying simulation methods to evaluate procedural sequencing in mediastinal staging. Our work extends this approach to the staging arena, focusing on the risk of false N2b upstaging when a single needle is used across multiple N2 stations.

This study was conceived to address an increasingly important but underexplored problem in thoracic oncology: the potential for cross-contamination during EBUS staging done without ROSE to spuriously upstage patients from single-station (N2a) to multistation (N2b) disease. Although the existence of contamination has been documented, the probability that it leads to a clinically meaningful change in stage is not known.

Our objective was to estimate the probability that a patient with true single-station N2 disease would be falsely classified as N2b if multiple N2 stations were sampled sequentially with the same needle and without ROSE. We also sought to estimate the number of additional needles needed to prevent one false upstage and to identify clinical contexts—central versus peripheral tumors, cN0 versus cN1 disease—where needle change may be justified. The goal is

to estimate a theoretical upper bound on false upstaging and not to assert the actual contamination likelihood.

METHODS

We developed a Monte Carlo decision-analytic model simulating mediastinal staging in the absence of ROSE in 100,000 virtual patients per scenario. A schematic of the decision structure for the high-risk central cN0 scenario is shown (Fig. 1). The simulation incorporated published data on the prevalence of occult N2 disease, the distribution of single-station versus multistation involvement based on published data, and estimates of contamination risk from the endosonography literature. Detailed model input parameters, sampling order assumptions, and sensitivity analysis structure are provided in the supplementary appendix (Table S1, Supplemental Digital Content 1, <http://links.lww.com/LBR/A407>). Patients were stratified by clinical node status (cN0 versus cN1) and tumor centrality (central versus peripheral).

Patients with pre-existing radiographic or PET-positive N2 disease were not included in the model. In such cases, bronchoscopists may adopt a different sampling strategy—progressing from the least suspicious to the most suspicious lymph node within the N2 station—to minimize the risk of carryover from a malignant to a benign node. Including such patients would therefore underestimate the true potential for cross-contamination in cases where disease is clinically unsuspected, which was the primary focus of this analysis.

For each simulated patient, the probability of occult N2 disease was set at 3% for peripheral cN0 tumors, 7% for central cN0 tumors, and 27% for cN1 disease.¹⁰ Among those with N2 disease, the base-case assumption was that 60% were single-station, with sensitivity analyses ranging from 50% to 75%.¹² Sampling order was modeled either as “least suspicious first,” in which the malignant station was sampled first in 20% to 30% of cases, or as “random,” with a malignant-first probability closer to 50%. Each station was sampled with a mean of 3 passes (range: 2 to 4).

Contamination was modeled at the per-pass level, with 3 priors informed by the literature: low (5%), moderate (15%), corresponding to catheter or channel contamination frequencies), and high (20%, reflecting a conservative interpretation of cytologic residuals in the EBUS setting).^{2,8,9} The full range of contamination scenarios and their interaction with $P(N2a|N2)$ values are summarized in the supplementary appendix (Table S3, Supplemental Digital Content 1, <http://links.lww.com/LBR/A407>). The cumulative probability of carryover from a malignant to a negative station increased with the number of passes. If contamination occurred, a true single-station N2 case was reclassified as N2b. The contamination probabilities were selected to represent a conservative upper-bound estimates rather than true EBUS-specific rates, acknowledging uncertainty and indirectness in the available literature.

Outcomes were calculated under 2 strategies: sampling multiple N2 nodes with a single needle versus changing the needle between N2 nodes (which eliminates contamination risk but incurs additional cost). Primary outcomes were the proportion of patients spuriously upstaged and the number of needles needed to prevent one false upstage (NNC). Lower NNC values indicate greater procedural efficiency. Computation details and probability distributions are presented in the supplementary appendix (Tables S2-S5,

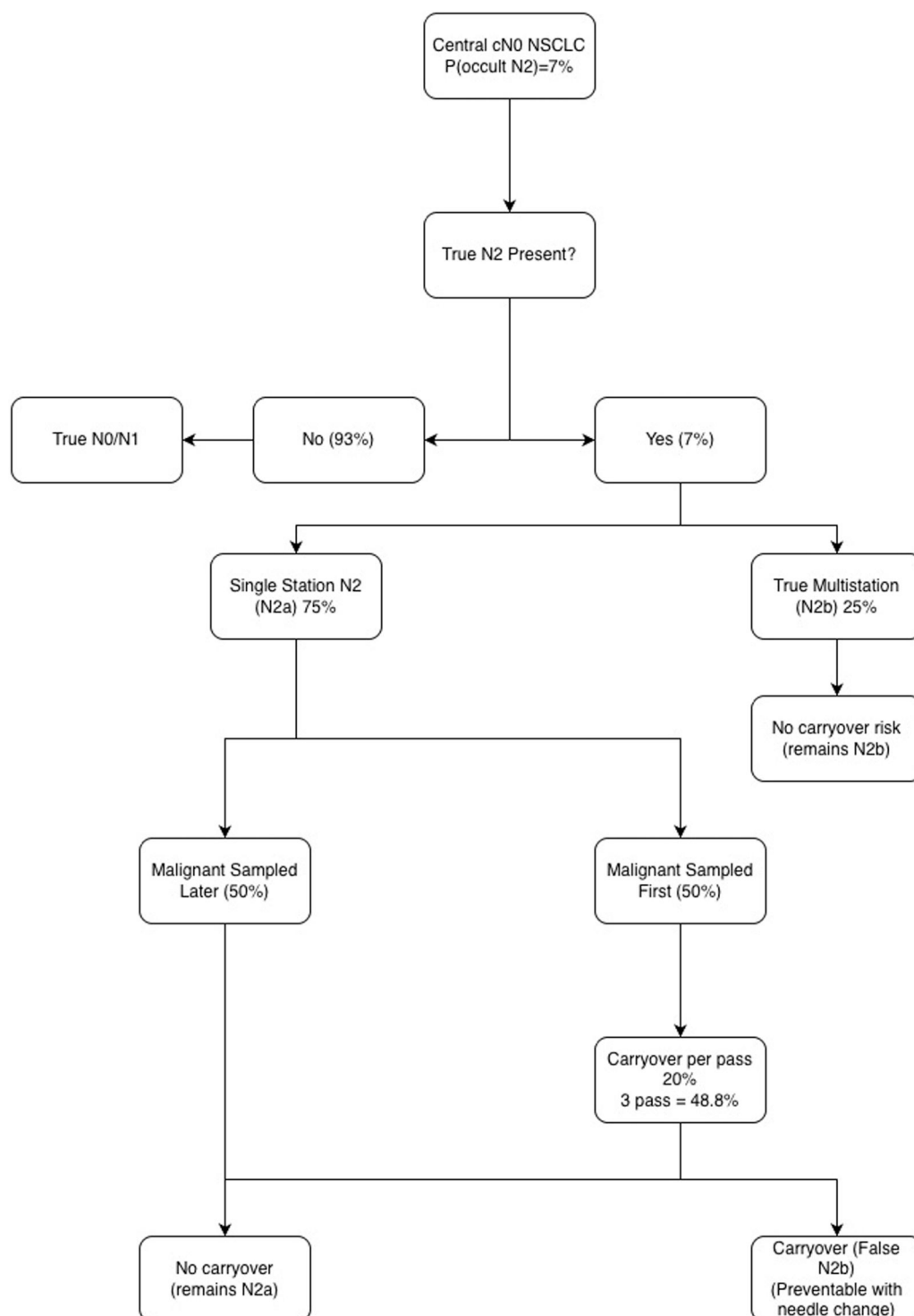


FIGURE 1. The Decision tree for the high-risk central cN0 scenario illustrates the sequence of node sampling, per-pass contamination events, and resulting classification (true or false N2a→N2b upstage).

TABLE 1. Base-Case Results By Scenario and Contamination Level

Scenario	Per-pass contamination (%)	P (N2a N2)	False N2a→N2b upstage %
Peripheral cN0	15	0.60	0.1-0.2
Central cN0	15	0.60	0.5-0.8
Central cN0 (high risk)	20	0.75	1.3-1.7
cN1	15	0.6	0.8-1.4

Supplemental Digital Content 1, <http://links.lww.com/LBR/A407> and Figure S3, Supplemental Digital Content 1, <http://links.lww.com/LBR/A407>).

RESULTS

The Monte Carlo simulation showed that false N2b upstaging during EBUS staging is possible but infrequent (Table 1). Complete numeric results across all modeled contamination rates and $P(N2a|N2)$ values are available in the supplementary appendix (Table S3, Supplemental Digital Content 1, <http://links.lww.com/LBR/A407>). The absolute probability of mis-upstaging is modest, reflecting the relatively low modeled rate of occult N2 disease in central cN0 tumors.

Under base-case assumptions, the probability of having single-station N2 disease given that N2 disease is present ($P[N2a|N2] = 0.60$, per-pass carryover $\approx 15\%$), $\sim 0.5\%$ to 0.8% of all patients with central cN0 disease were falsely upstaged from N2a to N2b when a single needle was used. The rate increased to about 1.3% to 1.7% in high-risk scenarios ($P[N2a|N2] = 0.75$, carryover $\approx 20\%$) but remained negligible in peripheral cN0 tumors ($< 0.2\%$). Patients with cN1 disease showed a higher probability, between 0.8% and 1.4% .

The efficiency of needle change was highly context-dependent (Fig. 2). In the base-case scenario, preventing one false upstage required roughly 200 to 350 additional needle changes in central cN0 disease and 100 to 200 in cN1 disease. In peripheral cN0 tumors, the number exceeded

1000. Even in the high-risk scenario, the number of needles required per false upstage avoided rarely fell below 150. Needle change efficiency across all scenarios is shown in Figure S2, Supplemental Digital Content 1, <http://links.lww.com/LBR/A407> and summarized in Table S4 in the Supplementary Appendix, Supplemental Digital Content 1, <http://links.lww.com/LBR/A407>.

Sensitivity analysis showed that variations in per-pass carryover probability, $P(N2a|N2)$, and sampling order each produced similar changes in model output. The tornado plot displayed comparable bar lengths for all 3 factors, indicating that uncertainty in false N2b estimation arises from multiple parameters rather than a single dominant driver. Detailed sensitivity analyses and tornado plots are shown in the Supplementary Appendix (Figures S3, Supplemental Digital Content 1, <http://links.lww.com/LBR/A407> and Table S3, Supplemental Digital Content 1, <http://links.lww.com/LBR/A407>).

DISCUSSION

This analysis suggests that in patients with N0/N1 disease, although cytologic contamination has been documented at the microscopic level, its translation into clinically meaningful mis-staging is rare. These findings appear to be consistent regardless of the scenario used. Detailed scenario analyses are presented in the supplementary appendix (Tables S2-S5, Supplemental Digital Content 1, <http://links.lww.com/LBR/A407>, Figures S1-S12, Supplemental Digital Content 1, <http://links.lww.com/LBR/A407>). Even in contamination-prone conditions, the probability of false N2b classification remains below 2% .

The results may suggest that, in the absence of ROSE, routine needle change between N2 stations may not be cost-effective in all case scenarios. In low-risk and moderate-risk scenarios, hundreds to thousands of additional needles would be required to avoid a single false upstage, representing a poor balance of benefit and procedural burden. This is particularly pertinent in resource-limited settings, where the cost and availability of needles can significantly constrain the capacity to perform EBUS for mediastinal staging. However, a selective approach—

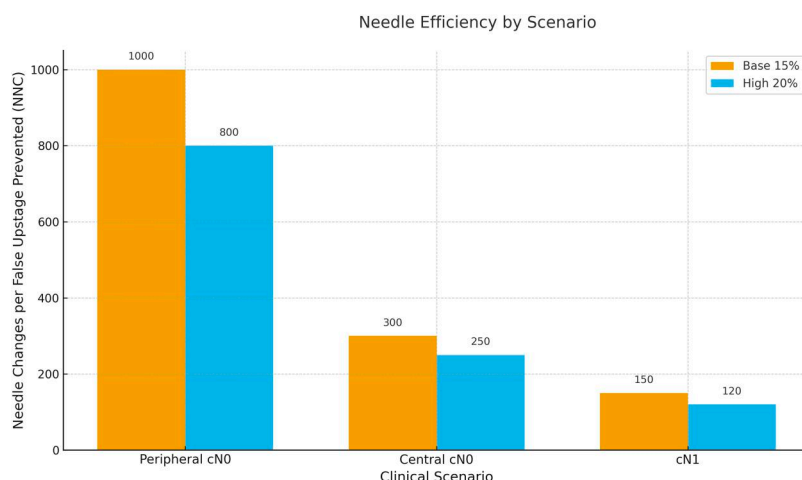


FIGURE 2. Needle efficiency by clinical scenario. The number of needle changes required to prevent one false upstage (NNC) is shown for peripheral cN0, central cN0, and cN1 disease at base (15%) and high (20%) contamination rates.

changing needles in high-risk contexts such as central or cN1 tumors with a higher pretest probability of single-station N2 disease—may be defensible in centers where contamination concerns outweigh incremental cost or where equipoise remains regarding borderline surgical candidacy.

Sensitivity analysis clarified that model uncertainty is multifactorial. Carryover probability, true $P(N2a|N2)$, and sampling order all exert similar influence on outcomes. This underscores the need for better real-world data across all 3 domains: standardized quantification of EBUS needle carryover, precise epidemiologic data on single-station versus multistation N2 disease, and procedural audits detailing actual sampling sequence “malignant vs benign lymph node first within the N2 lymph nodes.” The study therefore provides quantitative support for a risk-based, rather than universal, policy on EBUS needle reuse between N2 stations.

Although we did our best to utilize all the known data to try to estimate the potential risk of upstaging, our study carries several limitations: Several prediction models examined the prevalence of N2/N3 disease in patients with radiographic N0/N1 disease, each with slightly different variables. We used HAL and HOMER as it has demonstrated excellent discrimination with an area under the receiver operating characteristic curve (AUC) of 0.85 in the derivation cohort and 0.88 in external validation.^{10,17–20} Although the estimated risk of N2 disease may be slightly different between prediction models, the overall absolute effect on the model is likely low. While an argument can be made that tumor size may affect the likelihood of occult N2/N3 disease, HAL and HOMER models, which we utilized, did not use tumor size as a predictor of N2/N3 disease. We acknowledge that HAL and HOMER models estimate the probability of N2/N3 disease, not just N2 disease. Therefore, it is plausible that some of the patients we included in the model may have had N3 rather than N2 disease. However, this would further reduce the likelihood of cross-contamination between N2 lymph nodes, and increase the number of needle changes required to avoid one false upstaging. The estimates of the prevalence of N2a and N2b reflect pathologic N2 distributions and not strictly the subset of clinically occult N2 disease. Therefore, the true distribution among clinically unsuspected cases remains uncertain, and we incorporated this uncertainty into the model by using a wide sensitivity range (50% to 75% single-station).

We acknowledge that the actual cross-contamination risk is not known. Published data merely suggests that malignant cells are found in the needle and the sheath after sampling a malignant lymph node and do not confirm that malignancy would be diagnosed if a benign lymph node is subsequently sampled. The contamination probabilities used in the model were derived from heterogeneous endosonographic literature that may be different from routine mediastinal staging for N0 patients, which limits external validity and likely overestimates real-world contamination risk. Importantly, even under these conservative assumptions, the modeled probability of false N2b upstaging remained low. Ultrasound features during mediastinal staging may impact the decision as to which N2 lymph node would be sampled first.²¹ Estimating the effect of ultrasound findings would be difficult. In order to account for that, 2 case scenarios were adopted. One where the chance of sampling the malignant lymph node was the same

as a benign lymph node, and the other when the benign lymph node was sampled first in 75% of the time, which may reflect ultrasound findings altering the choice of which N2 lymph node is sampled first. In practice, adherence to a benign-appearing-first strategy within N2 stations, guided by ultrasound characteristics, may significantly reduce contamination risk without needing to change needles.

It is worth reemphasizing that the findings from this simulation is only relevant to centers where EBUS is done in the absence of ROSE. While ROSE is often viewed as a mitigation strategy for contamination-related misclassification, it is not without limitations. ROSE accuracy depends on cytopathology expertise, specimen quality, and interpretive context, and both false-negative and false-positive results have been reported.²² The simulation did not include the likelihood of cross-contamination affecting the eventual treatment decision. We acknowledge that not every N2a to N2b upstaging would result in a significant change in management. Estimating the effect of upstaging on the management plan is more challenging. Contemporary treatment decisions for N2 disease incorporate resectability, physiological reserve, response to induction therapy and institutional practice patterns. However, applying a probabilistic adjustment for management impact would further increase the NNC reducing the cost effectiveness of needle changing in the absence of ROSE.

CONCLUSIONS

Using contemporary epidemiology and a decision-analytic Monte Carlo framework, our simulation suggests that the risk of spurious N2a to N2b upstaging due to EBUS needle contamination in the absence of ROSE is small but not negligible. The effect is most relevant in central cN0 and cN1 tumors, where occult N2 disease is more prevalent, and essentially absent in peripheral cN0 disease. In the absence of ROSE, routine needle change between N2 stations provides minimal incremental benefit for most patients. A selective strategy—guided by tumor location, pretest probability, and procedural sequencing—offers a more rational compromise between diagnostic accuracy and procedural efficiency. Future prospective work should quantify EBUS-specific contamination rates, track real-world sampling order of N2 lymph nodes, and correlate cytologic carryover with actual stage migration.

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